

Find missing terms

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find the missing terms: 84, __, 62, __, 40. Copy or complete the first step.

2. Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

3. Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

4. Miro says: Miro treats a variable as a label instead of a number that can vary. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the missing terms: 84, __, 62, __, 40. Copy or complete the first step.
Answer 73 and 51.
2. Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer $C = 6n + 4$.
3. Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer 31.
4. Miro says: Miro treats a variable as a label instead of a number that can vary. Is this correct? Explain.
Answer State what the variable represents, substitute its value and preserve the operation structure.

Find missing terms

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Find the missing terms: 84, __, 62, __, 40.

6. Check this result using an inverse, estimate or known fact: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

2. Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

7. Prove or explain your reasoning: Find the first five terms of $4n - 1$.

3. Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

8. Identify and correct this misconception: Miro treats a variable as a label instead of a number that can vary.

4. Solve this independently and show every stage: Find the missing terms: 84, __, 62, __, 40.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

10. Write and solve a similar question to: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

Teacher answers and checking prompts

1. Find the missing terms: 84, __, 62, __, 40.
Answer 73 and 51.
2. Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer $C = 6n + 4$.
3. Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer 31.
4. Solve this independently and show every stage: Find the missing terms: 84, __, 62, __, 40.
Answer 73 and 51.
5. Use a second method or representation for: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer Expected result: $C = 6n + 4$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer Expected result: 31. A valid check must be shown.
7. Prove or explain your reasoning: Find the first five terms of $4n - 1$.
Answer 3, 7, 11, 15, 19. The explanation must show why the method works.
8. Identify and correct this misconception: Miro treats a variable as a label instead of a number that can vary.
Answer State what the variable represents, substitute its value and preserve the operation structure.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Find missing terms

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Find the first five terms of $4n - 1$.

6. Create a new example that tests the same idea as: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

2. Prove the answer to this question, rather than only calculating it: Find the missing terms: 84, __, 62, __, 40.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 31.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro treats a variable as a label instead of a number that can vary.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find the first five terms of $4n - 1$.
Answer 3, 7, 11, 15, 19. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Find the missing terms: 84, __, 62, __, 40.
Answer Expected result: 73 and 51. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer Expected result: $C = 6n + 4$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 31.
Answer One valid reconstruction is: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro treats a variable as a label instead of a number that can vary.
Answer State what the variable represents, substitute its value and preserve the operation structure.
6. Create a new example that tests the same idea as: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.